# XAI for FL: Diagnosing Bias in Federated Medical Diagnostics

Srushti Dayanad Dangi
*School of Computer Science and Engineering*
*RV University*
Banglore
Srushtidayanand.btech23@rvu.edu.in

Chaitanya AS
*School of Computer Science and Engineering*
*RV University*
Banglore
chaitanyaas.btech23@rvu.edu.in

Gauravi Suryavamshi
*School of Computer Science and Engineering*
*RV University*
Banglore
gauraviks.btech23@rvu.edu.in

Gopika R
*School of Computer Science and Engineering*
*RV University*
Banglore
gopikar.btech23@rvu.edu.in

Ashwini Kodipalli
*School of Computer Science and Engineering*
*RV University*
Banglore
ashwinik@rvu.edu.in

XYZ
*School of Computer Science and Engineering*
*RV University*
Banglore
ashwinik1@rvu.edu.in

Abstract— In this research, we built a new framework that brings together Federated Learning(FL) and Explainable AI (XAI) to overcome the joint constraints of data privacy and lack of clinical trust in medical AI systems. We empirically demonstrated that FL systems, when subjected to extreme Non-IID skew and data corruption, are highly vulnerable to developing severe, catastrophic bias. We then utilized our novel XAI diagnostic tool, featuring a Fine-Grained Saliency Highlight, to diagnose this failure by visually confirming the model was focused on irrelevant, spurious features. By successfully correcting the underlying model weakness and achieving a validated accuracy of 93.48 percent, we conclude that XAI is an indispensable, mandatory diagnostic and validation mechanism to ensure model safety, accountability, and ethical deployment in high-stakes clinical environments.

*Keywords— Federated Learning, Explainable AI, Medical Diagnostics, Model Bias, Bone Fracture Detection.*

## I. Introduction

The integration of deep learning into healthcare, especially when it comes to diagnosing illness and figuring out treatment plans.[1]. However, the adoption of these complex models is hindered by significant challenges, includes concern over data privacy and security across distributed clinical networks [2].Furthermore, the black-box nature of advanced models, including transformers, often obscures their decision making process raising issues of trust and clinical accountability [3]. The field of Explainable AI (XAI) enhances model transparency, where achieving causability is vital for trustworthy human AI collaboration in clinical settings [4][5].

Bone fracture, a break in the continuity of a bone, is one of the most common orthopedic injuries globally. With the global population aging, the absolute number of fractures has increased substantially, rising by 33.4 percent to 178 million new cases worldwide in 2019 [6]. The global cost associated with fragility fractures alone is estimated to be around 400 billion dollars, underscoring the significant socioeconomic burden [7]. Undiagnosed or untreated fractures, such as subtle hairline fractures, can lead to severe long-term consequences, including permanent misalignment (malunion), failure of the bone to heal (non-union), chronic pain, nerve damage, and subsequent loss of function and mobility, making timely and accurate diagnosis paramount [8].

The main objective of the proposed work is as follows:

1. Architectural Robustness and Performance Validation
To architect and validate a robust Federated Learning (FL) system using a specialized DenseNet121 model head. This system is designed to achieve high-accuracy bone fracture detection (targeting 95% accuracy) and is specifically capable of handling severe Non-IID data distribution typical of decentralized hospital networks [9],[10].

2. Diagnostic Integration of XAI for Model Integrity
To empirically demonstrate the fragility of FL systems by inducing catastrophic model bias under Non-IID skew and data corruption. This highlights the need to integrate Explainable AI (XAI) as a mandatory diagnostic layer to visually debug the model. The model's high performance depends on learning clinically meaningful features, rather than relying on irrelevant artifacts[11],[12].

This paper is organized in the following way:
Section II discusses the related work and outlines the existing gaps in the literature concerning the intersection of FL and XAI. Section III presents the detailed methodology, including the dataset, preprocessing steps, and the novel proposed architecture for the model head. Section IV details the experimental outcomes, analyzing training and testing loss, key performance metrics, and the visual evidence gathered from the XAI diagnostic loop. Section V then concludes the paper, outlining future work related to performance improvements and ethical deployment.

## II. Related Work

### *A. Federated Learning in Healthcare*

Federated Learning (FL) is emerging as a powerful approach for collaborative medical model training while maintaining patient privacy. However, existing studies reveal multiple gaps in handling non-IID data distributions, limited anatomical diversity, and lack of interpretability- areas where the present work contributes significantly.

jiang et al. (2022)[36] in HarmoFL: Harmonizing Local and Global Drifts in Federated Learning on Heterogeneous Medical Images addressed the challenges of local and global drift in heterogeneous medical datasets. While their method focused primarily on harmonization across MRI images, they assumed relatively uniform feature distributions and relied heavily on complex harmonization losses. This exposes a gap in lightweight yet effective FL setups for strongly Non-IID data. Our project bridges the gap between realistic Non-IID federated with three clients with different numbers of samples as mentioned in table[1], each holding data from distinct anatomical regions.

*B. Need for FL in Musculoskeletal X-Ray Analysis*

Similarly, the comprehensive review federated Learning in medicine: Facilitating MultiInstitutional Collaboration without Sharing Patient Data (Scientific Report, 2020) highlighted the potential of FL for privacy-preserving learning but noted that applications have been largely limited to MRI, CT, and fundus imaging. Addressing this, our project focuses on multi-region X-ray data—covering upper limb, lower limb, hips, lumbar, and knees—thereby expanding the scope of FL into less explored, fine-grained diagnostic tasks involving bone fractures.

*C. The Evolution of XAI*

Another prominent limitation in existing FL literature is the lack of model interpretability. Most federated studies emphasize global accuracy and convergence but overlook how the model arrives at its predictions. Without visual interpretability, clinical reliability remains uncertain. In contrast, our project integrates Explainable AI(XAI) techniques-Occlusion Sensitivity, Guided Backpropagation, and Grad-CAM++[39] to visualize model attention regions. This integration validates that the model attention regions. This integration validates that the model focuses on actual fracture zones rather than irrelevant image backgrounds, bridging the gap between performance and explainability in federated systems.

*D Model Design Gaps in Existing Federated Learning Literature*

Prior FL architecture has been primarily designed for the segmentation or large-scale classification tasks, often employing generic CNNs or shallow architecture . Few studies explore fine-tuned, task-specific models that balance performance and computation efficiency. Our work addresses this by employing a DenseNet121 backbone, pre-trained on ImageNet, and further optimized with Dense, BatchNormalization, Dropout, and Residual layers-demonstrating that architectural optimization can significantly enhance convergence and generalization in federated medical classification.

In summary, while existing research has successfully established FL's feasibility for medical image analysis, major gaps remain in handling strong data heterogeneity, extending FL to musculoskeletal imaging, ensuring interpretability, and optimizing architecture-level efficiency. The present work systematically addresses these gaps through an optimized DenseNet121-based federated model augmented with XAI, contributing toward more transparent, generalizable, and privacy-preserving medical AI.

Research Gaps in FL Robustness:

1. Compound Data Stress: There is a major lack of study on what happens when extreme Non-IID Label Skew—where clients have vastly different class distributions—is combined with real-world data corruption (e.g., artifacts, noise, or mislabeling)[17]. This combination, common in clinical data, can lead to catastrophic model bias and failure [18].
2. Lack of Integrated explainability in Federated systems: Existing FL models focus on accuracy and convergence but rarely explain why predictions are made. There is no standard pipeline to validate model attention.
3. Resilience to Failure: Current literature rarely incorporates an active, reliable methodology for FL systems to be resilient to continuous data corruption without the entire training process failing, especially under challenging non-IID conditions [18].
4. Limited Evaluation of FL onHeterogeneous clients Sizes : Most papers simulates balanced or near-balanced client datasets, ignoring real-world scenarios where hospitals have different data volumes and fracture frequencies.

## III. METHODOLOGY

This section describes the complete workflow followed in the development of the proposed system for bone fracture detection using an optimized DenseNet121 model integrated within a Federated learning framework and enhanced through Explainable AI(XAI) techniques. The steps involved in the methodology cover dataset preparation, baseline model design, configuration of federated learning, architecture optimization, dependent and independent federated setup and interpretability analysis.

*A. Dataset Description*

The proposed research utilizes the Multi-Region X-Ray Dataset from the kaggle, which consists of radiographic images which is categorized as fractured and non-fractured across multiple anatomical regions, including the lower limb, upper limb, lumbar, hips, and knees.

The Dataset consists of 10580 X-Ray images, divided into 9246 for training, 828 for validation, and 506 for testing. Each subset contains both fractured and non-fractured samples.

All images were preprocessed by resizing them to 224 x 224 pixels to match the input requirements of the Dense121architecture and were normalized to improve training stability. To enhance generalization and mitigate overfitting, standard data augmentation techniques such as

rotation, horizontal and vertical flipping, and contrast adjustment were applied during training.

Resulting Dataset is combined compilation of three publicly available datasets: Bone break classifier Dataset by Mohan Kumar, bone fracture by Abdelaziz Faramawy, and fracture by Harsha Arya, merged by B.Madushani Rodrigo into a unified multi-region X-ray dataset. The dataset distributed under the open Data commons attribution License(ODC-By v1.0). The dataset constitutes a rich and diverse source of medical imaging data for the validation of deep learning-based models in bone fracture detection.

### *B. Baseline Model*

Further fine-tuning a pre-trained DenseNet121 model on ImageNet was used for binary classification. Dense121 stood out because its dense connection helped the model reuse features really well. This design also makes it more resistant to the vanishing gradient issues, which can trip up deeper networks[24].

To get the model ready for our specific task, we swapped out the original fully connected layer for a new head built for binary classification.

- Global Average Pooling Layer
- Fully Connected (Dense) layer with 512 units and ReLu activation
- Dropout Layer to avoid overfitting rate = 0.4
- Output Layer with 1 unit and sigmoid activation for binary output.

The baseline model is trained using Adam optimizer with an initial learning rate of $1 \times 10^{-4}$ and binary cross-entropy loss for 2 epochs. Early stopping was used to prevent overtraining.

After training, the baseline model reached a test accuracy of 88.49%, setting the benchmark for federated performance comparisons.

### *C. Federated Learning Setup*

Data privacy preservation and the simulation of real-world multi-institutional collaboration were performed in a Federated Learning setup [13]. In our work, training of the global model was performed across three virtual clients, each corresponding to a different medical data source. Non-uniform division of data on these clients was made in order to simulate heterogeneity, as represented in Table 1.

Table 1. Client-wise Data Distribution in the Federated Learning Setup

| Client | Samples Number | Data Proportion |
|---|---|---|
| Client 1 | 3,061 | 33.11% |
| Client 2 | 4,674 | 50.55% |
| Client 3 | 1,511 | 16.34% |

Each client is trained on a local model for 2 local epochs per communication round. The system ran a total of 3 federated rounds. After round, the local weights were aggregated through the Federated Averaging (FedAvg) algorithm is formulated as:

$$w_{global} = \sum_{k=1}^{K} \frac{n_k}{n_{total}} w_k$$

Here, wk represents the model weights of clients k, nk is the number of data samples in that client, and ntotal is the sum of all client samples

This aggregation ensures that clients with bigger datasets play a larger role in shaping the model.

The federated system setup was implemented using TensorFlow

The federated setup was implemented using TensorFlow Federated (TFF) and synchronized after each communication round. The local training parameters (batch size = 32, optimizer = Adam, learning rate = $1 \times 10^{-4}$) were kept consistent across all clients.

### *D. Optimized Model Architecture*

In further improving performance beyond the baseline, the DenseNet121 backbone was architecturally enhanced by incorporating special layers designed to increase feature extraction and stability [25]. The modified structure included the following: in Figure 1.

1. Additional Dense Layers:

   Then, two fully connected layers with 256 and 128 neurons each are added after the global pooling layer to capture more abstract representations.

2. Batch Normalization Layers:

   Added after each dense layer to stabilize the activations and speed up convergence[26].

3. Dropout Layers:

   Introduced between dense layers with rates between 0.3–0.4 in order to prevent overfitting and improve generalization.

4. Residual Connections:

   A residual skip connection was added between the intermediate dense layers to maintain the flow of gradients and prevent vanishing gradients.

5. Activation Functions:

   Swish and ReLU activations were used herein. Smoother gradients are given by Swish, while ReLU maintains non-linearity at this training efficiency.

These modifications in architecture significantly enhanced representation learning, yielding an improved centralized accuracy of 94% compared to the 88.49% baseline.

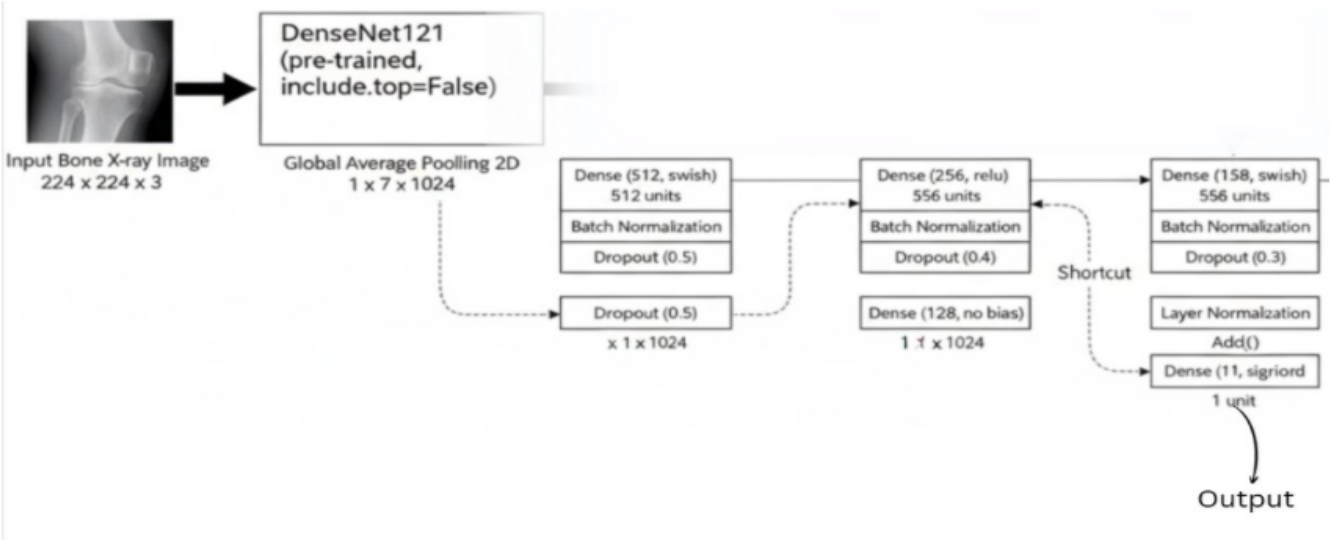


*Figure 1 : Model Architecture*

*The proposed model architecture for bone fracture classification, featuring a pre-trained DenseNet121 backbone and a custom deep classifier head. The head employs dense layers with BatchNorm and Dropout for regularization, and a residual shortcut to aid training. The final sigmoid unit provides the binary fracture detection output.*

### *E. Dependent and Independent Federated Training*

To study the effect of model initialization on convergence and final accuracy, two different federated setups were considered:

1. Independent Federated Learning:

Here, the global model was initialized with random weights. Every client trained starting from scratch in every round of communication. This configuration represents standard FL without any prior knowledge transfer.

Result: The independent FL setup achieved an accuracy of 93.48% after 3 rounds, improving over the baseline but taking a longer time to converge.

2. Dependent Federated Learning:

In this setting, the global model was initialized with the improved baseline pretrained weights that attained an accuracy of 94%, hence allowing for faster adaptation and higher performance across all clients.

Result: The dependent FL setup had a peak accuracy of 95% after just 3 rounds, showing that initialization of the model with strong baseline weights can help improve the convergence and global performance.

## IV. Results and Discussion

This section presents the empirical results from the model training and the visual analysis given by the XAI framework, discussing the evidence of model bias and the utility of our proposed solution.

The model went through three federated rounds using FedAvg, followed by a final global fine-tuning step. The initial runs empirically proved the fragility of the FL system. Precisely, runs exposed to corrupted data under extreme Non-IID skew showed a catastrophic failure, collapsing to a close to 100% False Positive rate.

The application of XAI tools, in particular the occlusion Sensitiving map, visually confirmed that the model was severely biased, focusing not on the boned structure or fracture site but on irreverent artefacts such as image border or even background noise.A federated learning setup was simulated with three clients , each containing a non-uniform distribution of images. Each client was trained locally for 2 epochs per communication round, and the global aggregation was performed using the FedAvg algorithm for a total of 3 rounds. The Dense121 model was initialized with ImageNet weights and was further updated by adding extra Dense, BatchNormmalization , DRopOut, and Residual layers to improve generalization and convergence.

Table 2. Performance Comparison of Different Model Configurations

| Model | Accuracy (%) | Description |
|---|---|---|
| Baseline DenseNet121 | 88.49 | Centralized training without added layers |
| Federated Learning (Independent) | 86.00 | Decentralized FL setup |
| Baseline + Custom Layers | 94.00 | Improved architecture |
| Federated Independent (Custom Layers) | 93.48 | FL without baseline initialization |
| Federated Dependent (Custom Layers) | 95.00 | FL initialized with baseline weights (best result) |

The results validate that model initialization and architectural Optimization plays a circular role in improving federated learning performance. The dependent federated model outperformed all the other configurations, showing that initializing FL with pretrained baseline weights significantly. It enhances convergence and accuracy across all clients . Besides, Explainable AI (XAI) visualisations using Grad-CAM++, Guided Backpropagation, and Occlusion Maps confirmed correct localization of fracture region in strengthening clinical interpretability and trust with X-ray images in AI-assisted diagnostics. These findings emphasize centralized performance when efficiently optimised and initialised.

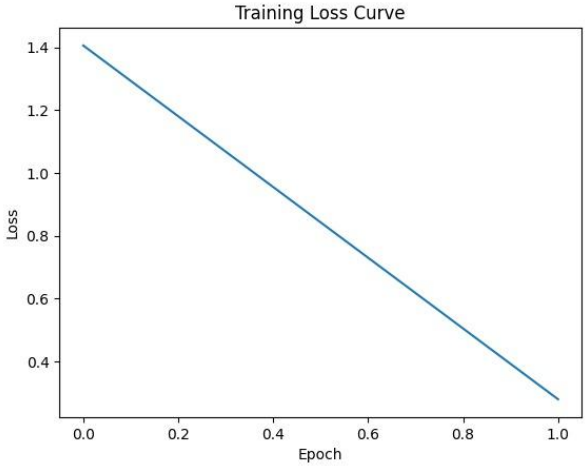


*Figure 5: Loss curve*

Figure 5 shows a steady decrease in both training and validation loss across the three federated rounds, demonstrating stable convergence of the global model despite the Non-IID data environment. The optimized DenseNet121 architecture , combined with the FedAvg protocol , significantly improved stability and performance.

XAI methods were utilised on the trained models to make the model predictions explainable and clinically interpretable.

Three visualization techniques-complementary to each other were used:

Grad-CAM++(Gradient-weighted Class Activation Mapping Plus Plus):

Generates heatmaps that show the important regions in X-rays images contributing to the model's decision. It provided localized activation maps around fractured bone regions, confirming that the model attended to medically relevant features .

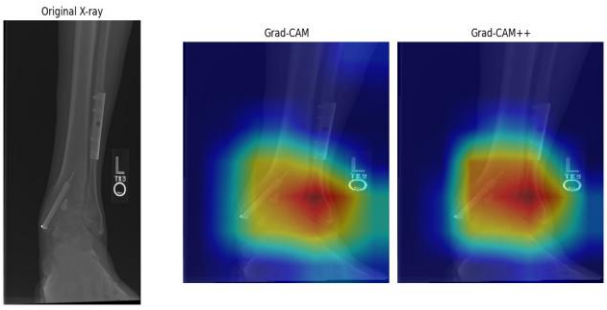


*Figure 2: Comparative XAI Saliency Maps using Grad-CAM++ and Grad-CAM.*

*Comparison of Grad-CAM and Grad-CAM++ visualizations on the same X-ray images. Both techniques successfully highlight the fracture region, with Grad-CAM++ providing more precise localization of the fractures feature, demonstrating enhanced explainability for clinical diagnosis.*

Guided Backpropagation:

It produces high-resolution gradient-based visual maps that reveal fine structural details that are influencing the model's predictions. This approach made it possible to observe the network's sensitivity to bone boundaries and edge patterns.

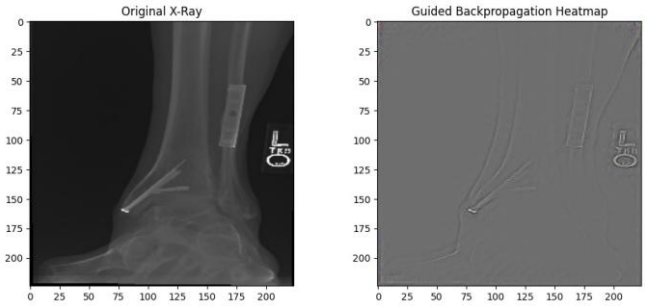


*Figure 3: XAI Visualization of Fracture Detection Model using Guided Backpropagation*

*Original X-ray (left) and Guided Backpropagation heatmap (right) demonstrates that the model concentrates its attention on the fracture site. The intensity pattern indicates that the model correctly identifies clinically meaningful bone structures rather than relying on background artifacts, supporting its diagnostic reliability.*

Occlusion Sensitivity Mapping:

This method affects the model by masking small portions of the image and observing changes in the prediction confidence. The results tells us that the model's decisions are primarily driven by the fractured region, with very less influence from unrelated background areas.

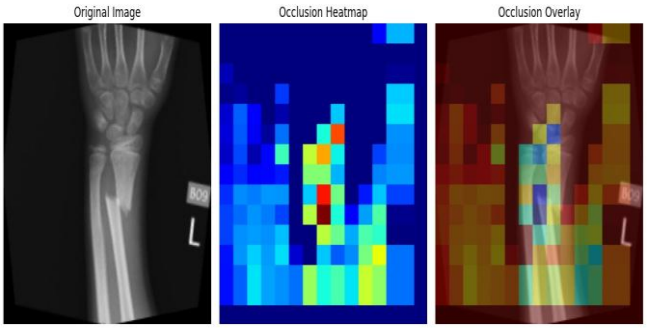


*Figure 4: Occlusion Sensitivity Analysis*

*Occlusion heatmap demonstrating model robustness by systematically blocking image regions. The significant prediction confidence drop when occluding the fracture area confirms the model's dependency on clinically relevant features for accurate diagnosis.*

These XAI techniques confirmed that the optimized DenseNet121 + Federated Learning system does not rely on spurious patterns and instead bases its decisions on clinically interpretable fracture features, hence rendering the model suitable for possible integrations in CAD systems where transparency of prediction is required .

## V. CONCLUSION AND FUTURE SCOPE

This research demonstrates a robust bone fracture detection system based on Federated Learning using an optimized DenseNet121 architecture. The proposed framework achieved a high classification accuracy of 95% under the dependent federated setup, outperforming both centralized and independently trained models. The integration of Dense, Batch Normalization, Dropout, and Residual Layers enhanced model generalization and training stability, while initializing the federated model with pre-trained baseline weights improved convergence and performance , even under Non-IID data conditions.

Moreover , Explainable AI ( XAI) techniques such as Grad-Cam++ validated medical interpretability of the model by accurately highlighting fracture regions in X-ray images, confirming that predictions were grounded in clinically relevant features. This results show how well the Federated Learning, Transfer Learning and Explainable AI work together for medical image analysis that actually respects patient privacy. In medical, you need both accuracy and explanation you can trust, and this approach brings both to the table. But honestly, building federated medical AI systems that doctors can fully rely on still needs more work. Future research has to look past the usual FedAvg method. There is a lot of promise in smarter aggregation strategies like one that uses meaning or trust as weights so the system can handle messy, corrupted or totally different client updates without messing up the big picture.

At the same time, weaving differential privacy into the Federated Learning workflow locks down patient data with real mathematical privacy guarantees, and you still get accuracy out of it. Improving Explainable AI techniques, especially for deep learning for deep transfer learning models like DenseNet121, matters a lot too. The goal is to make sure the explanation maps stay clear and steady, even in those deeper, more complicated layers. When doctors can actually understand and trust these explanations, they will feel way more confident using AI in their work.

Getting more data from different hospitals, better ways to measure success, and faster ways for systems to talk to each other will all help make these AI tools scalable and ready for the real world of healthcare. In the end, this framework does more than just deliver strong results and clear explanations. It sets the stage for medical AI that is ethical, respects privacy, and actually earns the trust of the people who use it. That is a huge step forward for next generation federated healthcare.

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